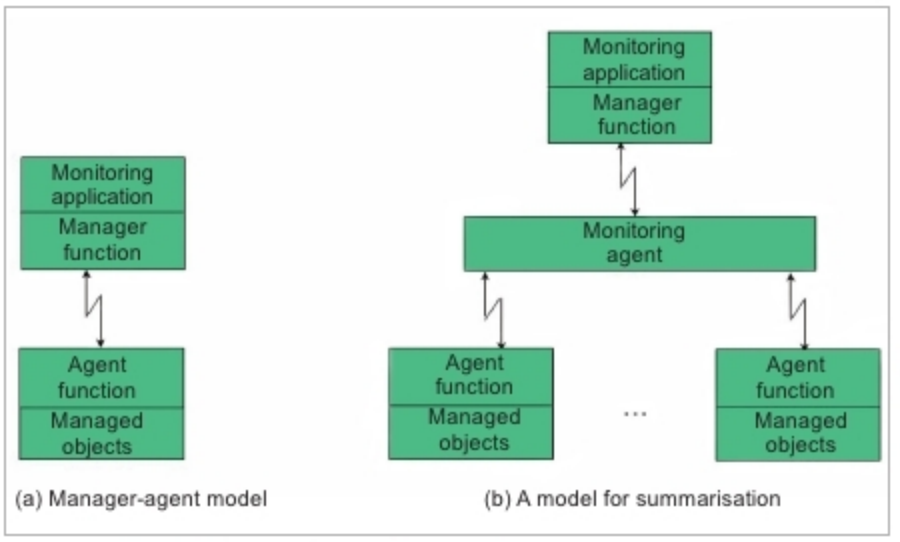


Figure 1: Functional architecture of a network monitoring system (Image source: *googleimages.com*)

attributes include node availability, circuit availability, etc. The time frame used for monitoring performance must be long enough to establish a network model. All the information extracted through performance monitoring is basically used to plan future network expansion and hence locate current network usage problems.

2. **Fault monitoring:** This deals with detecting the problems or issues in the network. It covers the different layers of the network since a problem can occur in any one of them. It requires establishing the 'normal' characteristics for the network over an extended time period. There are always some errors in the network but this does not mean that the network has persistent problems. Some of the errors expected to occur include noise present in a network link that can lead to transmission errors. The network gets into major problems when the number of errors suddenly increases above its normal behaviour. Hence, a record of normal behaviour is quite important.
3. **Account monitoring**: This deals with how any person uses a specific network. The network keeps a complete record of all the devices on it, which are used by people and also how often they are used. This type of information is generally used for billing users for the network usage and also for predicting future network usage.

The need for network monitoring

Network monitoring continues to be crucial for different businesses—to monitor their networks in order to be more productive and to avoid possible threats due to network failures and server downtime. Here is why we really need network monitoring.

1. Network monitoring analyses the performance of a network in real-time. So if an issue is detected, network administrators can be immediately alerted about it. This means that we can be informed about network problems wherever we may be, allowing for instant corrective measures and hence minimising potential downtime.
2. To measure the response time and consistency of a network and accordingly extract reliability metrics for them.
3. To validate availability and uptime for the specified network.
4. To diagnose different issues observed in the network by monitoring its behaviour and comparing that with the usual trend. It also helps in generating data for trend analysis creating a performance benchmark for the network. It actually generates a well-structured report, which can help us identify trends and patterns in a system's performance.
5. To allow network management applications to check the state of various parts of the network and the network devices present at remote locations. This can assist us in identifying specific areas of the network that are experiencing problems.
6. To collect useful information from different parts of the network so that the same set of information can be used to manage and control the network, going forward.
7. To keep an eye on the network traffic and bandwidth usage. It can validate whether crucial network components like routers, switches and servers, are down or up and running.
8. Some of the network monitoring packages also handle service-level agreements (SLAs) and quality of service (QoS) monitoring.

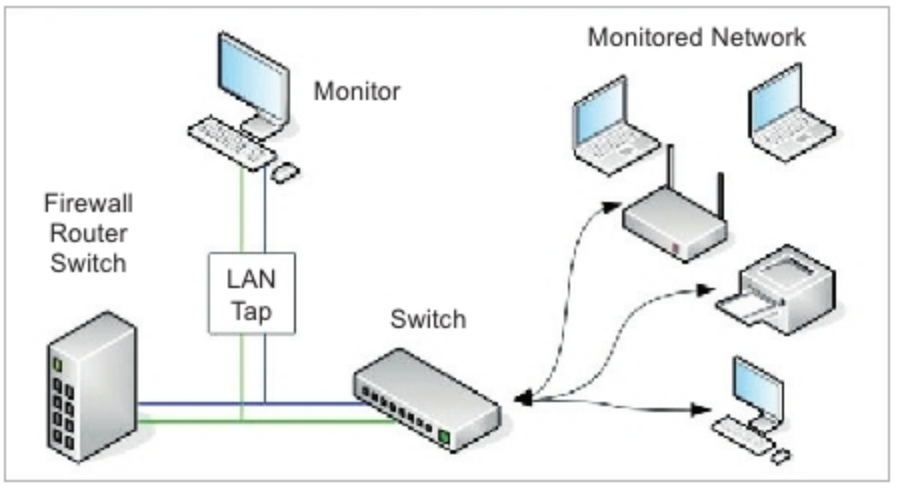


Figure 2: Basic workflow for a network monitoring system (Image source: *googleimages.com*)

Different network monitoring methods

Network monitoring involves a wide variety of methods that are implemented by IT professionals to maintain the integrity and security of an internal network.

1. **Packet sniffing:** This is a network monitoring technique that inspects every packet of information that passes through the specific network. Packet sniffers can detect different unauthorised network monitoring software, which might have been installed by hackers for spying on various business activities and information processes.
2. **Intrusion detection:** Intrusion detection monitors different local area networks for any unauthorised access by hackers. It can be implemented manually, though a majority of IT professionals prefer intrusion detection programs that automatically detect malware and viruses, network vulnerabilities such as logic bombs, backdoors and other security threats. This detection system is also used by individual computer systems that are connected to the network and file settings.